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Penn State researchers programmed AI models and trained computer vision system to track plant growth

Timothy Bueno(<https://www.fertilizerdaily.com/author/timbueno/>) | March 3, 2025, 6:00 pm | 103

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Researchers at Penn State have developed an automated computer vision system to enhance the monitoring of specialty crops grown in soilless systems within controlled environment agriculture (CEA). This innovative system allows for the continuous and precise tracking of plant growth, promising significant advancements in the year-round production of high-quality specialty crops. (<https://www.fertilizerdaily.com/>)

The research, conducted by an interdisciplinary team, underscores the need for precision agriculture technologies in competitive and sustainable farming. The system integrates artificial intelligence (AI) with an “Internet of Things” (IoT) framework, enabling detailed and frequent data collection essential for effective crop management. The findings of this study are detailed in the journal *Computers and Electronics in Agriculture* (<https://www.sciencedirect.com/science/article/abs/pii/S0168169924012572?via%3Dihub>).

Team lead Long He, an associate professor of agricultural and biological engineering, emphasized the traditional challenges of crop monitoring in soilless CEA systems, which typically require intensive labor and are limited in the frequency of data collection. He stated, “Automated crop-monitoring systems allow continuous monitoring of the plants with frequent data collection and a more efficient and informed management of the crop.”



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The core innovation of the study is the use of a recursive image segmentation model. This model processes high-resolution sequential images taken at predetermined intervals, accurately tracking changes in plant growth. The researchers demonstrated this capability by monitoring baby bok choy, highlighting its potential applicability to various crops.

Dr. Chenchen Kang, a postdoctoral scholar in He's lab and the study's first author, was instrumental in programming the AI models and refining the computer vision system to achieve robust performance throughout the crop growth cycle. Kang's work involved installing sensors, collecting and processing data, and developing the methodology alongside coding the AI models.

Francesco Di Gioia, associate professor of vegetable crop science and principal investigator on a related federal project titled "Advancing the Sustainability of Indoor Urban Agricultural Systems," stressed the importance of integrating various expertise to enhance the efficiency and sustainability of CEA systems. "The ability to automatically monitor and collect data on the crop status, estimate plant growth and crop requirements, along with the monitoring of the nutrient solution and environmental factors, will revolutionize how we manage crops," said Di Gioia.

Looking forward, Di Gioia anticipates that the integration of such precision agriculture technologies could enhance the quality of specialty crops and tailor their nutritional profiles to better meet consumer needs.

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